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Diversified flood governance and related socio-spatial vulnerability in Tana River County, Kenya

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Abstract

What are the consequential socio-spatial vulnerabilities of fragmentation in flood risk governance (FRG) functions in Tana River County, Kenya? To answer the question, this paper utilises social differentiation theory to establish types and consequences of such fragmentation. It applies theoretical patterns and processes of social differentiation to expose that FRG diversification could result in fragmentation of FRG functions. The study employs an interview strategy that focuses on governmental and non-governmental FRG actors. The results reveal that fragmented FRG functions in Tana River County was not an intentional policy decision, but rather a product of a transitional vacuum between policy promulgation and implementation. Furthermore, socio-spatial vulnerabilities differentiated by patterns and processes related to age, gender, disability, and ethnicity, for instance, are consequential outcomes of fragmented FRG functions in the Tana River floodplains. The study recommends implementation of the National Disaster Management Policy in Tana River County and institutionalisation of a specific FRG policy and legal framework.

KEYWORDS

differentiation, diversification, flood risk, fragmentation, governance, vulnerability

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1 | INTRODUCTION

Floods are a frequent cause of massive devastation (Jonkman, 2005; Nygren, 2016). People choose or have no other choice but to live in dynamic coastal zones and floodplains in pursuit of the ecosystem services therein, which fuel their livelihood activities (Imeson, 2012). People who live in coastal floodplains are at risk of the impacts of climate change and other economic developments that affect riverine and tidal flooding (Shampa et al., 2021). While the flood source is the river channel, to cause devastation, characteristics of overflow, such as speed of onset, magnitude, spatial extent, and depth, must interact with human settlements exposed and vulnerable to such features (Franke, 2004; Wisner et al., 2014; Montz, Tobin, and Hagelman, III, 2017).

Therefore, in the face of climate change, flood risk governance (FRG) is not only a matter that flood risk management (FRM) actors and decision-making processes must address to reduce or mitigate the impacts of flooding. In addition, it also poses a challenge to adaptation and transformation, because it concerns overseeing FRM strategies for actions regarding flood defence, prevention, mitigation, protection, and recovery (Gralepois, 2020). FRG includes actor networks, discourses, and multi-level coordination mechanisms in the pursuit of FRM (Alexander, Doorn, and Priest, 2018). The classical or traditional FRG pattern centres on a prevention-oriented approach, focusing on infrastructural flood defences such as dams and dikes (De Bruijn and Klijn, 2001; van Popering-Verkerk and van Buuren, 2017). The classical or traditional FRG pattern concentrates on infrastructure. An alternative integrated approach has its foundations in integrated watershed management, especially for arid and semi-arid areas, such as the Tana River County in Kenya, where flexibility is required to manage water scarcity in periods of drought and surpluses in times of floods (Bouwer, 2002). Diversification of FRG strategies is widely regarded as a necessary step forward to lessen the likelihood and magnitude of flooding, as well as to minimise the exposure of people and property, and in turn, the disruption, economic damage, health impacts, and other adverse consequences that ensue when floods occur (Gilissen et al., 2016). However, despite the political and instrumental efforts associated with the diversification of FRG, the ramifications of floods are still devastating (Gralepois, 2020).

Not all diversification of FRG affects socio-spatial vulnerability. However, a likely and inevitable outcome of diversification is that it can lead to fragmentation within FRG function areas and strategies, which can prove detrimental to the effectiveness of FRG (Gilissen et al., 2016). Hence, in many diversified strategies, FRG turns into a matter of adaptation and self-responsibilisation, which does not account for the socio-spatial distribution of vulnerabilities (Nygren, 2016).

By overlooking the socio-spatial distribution of vulnerabilities, fragmentation in FRG strategies could create social inequalities. Existing scholarship on social inequalities links differentiation to social hierarchisation, and difference to social inequalities (Lee and Juteau, 2003, pp. 8, 10). Hierarchisation in this case is the distribution of responsibilities between actors and governing rules enacted within different policy domains in FRG, which affect social inequalities. Lee and Juteau (2003) state that to understand the processes of social differentiation and its outcomes, social differences need to be comprehended and explained.

Apropos of flood protection governance within an institutionally fragmented environment, the development of public-private financing and implementation mechanisms for flood prevention and preparedness are differentiated by functions and areas, and the degree to which safety or risk of flooding is considered as optimal (Broekmans and Correljé, 2008). A novel finding is that social differentiation in flood impacts is relatively small soon after a flood, but widens over time, with socially disadvantaged groups enjoying less recovery (Houston et al., 2021). Thus, in floodplains such as the Tana River Delta, where decisions concerning financing and implementation of FRM mainly focus on the emergency (Okoko, 2022b), socio-spatial vulnerabilities owing to long-term flood impacts are easily overlooked in FRG and therefore continue to worsen with each flood experience. In addition, changes in flooding resulting from climate change impacts may also alter the spatial differentiation of vulnerability depending on the sensitivities and adaptive capacities of groups inhabiting floodplains (Coulily et al., 2015). The coastal Tana River Delta continues to face seaward global warming impacts, such as increased coastal erosion and salinisation, as well as

hinterland climate change impacts, such as flash floods followed by long spells of drought (Kairu, 1997; Opere, 2013; Gitau et al., 2021).

This paper uses Tana River County, Kenya, as a case to contribute knowledge on the consequences of diversified FRG and resultant fragmentation for socio-spatial vulnerabilities. It aims to explore how fragmentation in FRG for the Tana River catchment affects socio-spatial inequalities, which reflect in socio-spatial vulnerability to floods. To do so, this paper triangulates results from key informant interviews conducted within state and non-state agencies in the Tana Delta Sub-County with a document review of existing FRM policies in Kenya, including Tana River County. The paper then utilises social differentiation theory to establish the types and consequences of fragmentation in FRG in Tana River County. Ultimately, it seeks to answer the following research question: 'What are the consequential socio-spatial vulnerabilities of fragmentation in FRG functions in Tana River County, Kenya?'

1.1 | Theoretical background

Priority 2 of the Sendai Framework for Disaster Risk Reduction (SFDRR) 2015–2030 articulates strengthening disaster risk governance to manage disaster risk.¹ The SFDRR emphasises the importance of mainstreaming and integrating disaster risk reduction (DRR) within and across all sectors to review and promote coherence—coherence in disaster risk governance, however, was a pre-existing challenge in African countries (Sorell, 2013). While the SFDRR calls for integrated FRG, diversification in FRG strategies is a European Union policy and project concept focused on flood resilience, seeking to distinguish strategies of prevention, protection, preparedness, emergency response, and recovery (Oosterberg, Drimmelen, and Van der Vlist, 2005; Hegger et al., 2013). A major problem with the diversification of FRG strategies is that each way of doing so will not be feasible everywhere (Hegger et al., 2014). Diversification of FRG may also imply that new or more actors acquire a role in FRM, adding complexities and potential conflicts of interest (Broekmans and Correljé, 2008; Meijerink and Dicke, 2008; Hegger et al., 2014). Therefore, FRG actors have distinct roles and interests, as well as different policy issues of concern in terms of contributions to the resilience of the floodplain. Actors may differ in their interpretation of what socio-spatial vulnerabilities there are and how they should address them, which may hinder any changes at all that would benefit groups and individuals living in and depending on the floodplain.

Governance consists of the ways in which multiple and different actors behave, operate, interact, and confront each other in relation to the decision-making process on public affairs, which includes risks and disasters (Forino, Bonati, and Calandra, 2018). The concept of fragmentation in governance concerns the extent to which the world is composed of distinct clusters or divisions, regional or sectoral, of governmental organisation-based cooperation (Greenhill and Lupu, 2017). These distinct divisions attend to distinct functions, such as FRG functions. Therefore, they facilitate different decision-making processes and either shared or specialised responsibilities pertaining to FRM. These responsibilities concern specific FRM strategy actions and thus require specific competencies. Fragmentation in this case goes beyond institutions and their FRG functions to encompass their responsibilities and individual and/or holistic competencies for tackling FRM.

Flood risk is a regional governance issue because it happens in geopolitical regions, countries such as Kenya, and in specific sections of such regions, including river catchments such as the Tana River. It is also a sectoral governance issue because flood impacts affect sectors such as food and agriculture, health, energy, housing, wildlife, trade and commerce, education, and transportation. When FRG functions are integrated, there is central coordination regionally and between sectors and strategies (Gilissen et al., 2016). When FRG functions are diversified, regional and sectoral authorities, public, private, and non-profit, must cooperate, each bringing their own contributions and interests to FRM strategies (Hegger et al., 2014; Gilissen et al., 2016). Diversification in FRG functions thus translates into diversification in FRM strategies. In some cases, the focus will shift to what involved actors must do, and how they must do it. This shift depends on their institutional structures and processes, which alienates them from the recipients of FRM and the residents of floodplains, as well as their socio-spatial vulnerabilities (Nygren, 2016).

Such diversification in FRG functions could produce the unintentional consequence of fragmentation in FRG, reproducing socio-spatial inequalities, which in turn link with the socio-spatial vulnerabilities of floodplain inhabitants.

Gilissen et al. (2016) identify four main situations of fragmentation in FRG in Europe:

- Type 1 fragmentation refers to situations in which distinct actors operating in different policy domains pursue different FRM strategies. The entire existence of FRG is to ensure that the strategies in different policy areas for FRM work together effectively to reduce vulnerability and manage risk. However, when each policy domain has different actors and they act autonomously or semi-autonomously without a central coordinating agency, harmonisation of FRM outputs may be difficult to coordinate and achieve.
- Type 2 fragmentation refers to situations in which distinct actors operating in the same policy domain pursue the same FRM strategy. This scenario presents a situation where FRG actors address the same policy issue; with poor coordination or communication there could be poor implementation of strategies because each actor would think that the other should be responsible.
- Type 3 fragmentation refers to situations in which distinct actors operating in different policy domains pursue the same FRM strategy. This has similar consequences to the one above except that it will involve specific activities with regard to FRM strategies.
- Type 4 fragmentation refers to situations in which distinct actors operating in the same policy domain pursue different FRM strategies. In this case, there could be poor harmonisation between strategies that are dependent on each other, or delays, owing to financing or resource allocation for one activity that could result in delays in other activities.

Gilissen et al. (2016) depict fragmentation in the European case as differentiated by policy, actor, and FRM strategy implementation. There are no similar studies presenting a Kenyan case. Policies, actors, and strategy implementation are all components of society that hinge on multidimensional political, economic, cultural, and normative processes, which may or may not be affected by functional hierarchisation (Lee and Juteau, 2003). Understanding the patterns, categorisations, and hierarchisation behind the political, economic, cultural, and normative processes of FRG fragmentation could be essential in explaining how socio-spatial inequalities and, in turn, vulnerabilities can be traced to applying diversified FRG in floodplains, where it may not be feasible. Patterns, categorisations, and hierarchisation are concepts in the theory of social differentiation.

Social differentiation theory assumes that societies internally are highly diverse and differentiated, and differences proceed from the more general to the more special (Albert, 2016, p. 48). The theory in its original form views functional differentiation, which equates to fragmentation, as a positive step in development. Fragmentation allows the more specialised functional units to make changes and decisions easily concerning their functions and strategies—functional differentiation has had proven success in solving former fundamental social problems (Albert, 2016). Lee and Juteau (2003) underlined that previous arguments against functional differentiation in the application and development of the theory by scholars such as Spencer (1887), Durkheim (1893), Parsons (1985), Luhmann (1982, 1995), and Alexander and Colomy (1990) were incomplete as they did not answer questions regarding how social categories manifest or their construction. Lee and Juteau (2003) argued that social groups are interrelated, with majorities and minorities socially constituted through hierarchisation and unequal life chances.

In disaster research, scholars such as Clausen (1992) suggested that to explain disasters sociologically, one ought to start with social differentiation theory, because the most dangerous feature making a society prone to such events is its social differentiation between problem-solvers and problem laymen. Tompkins and Adger (2004) pointed out that climate change impacts, such as increased flooding, are spatially and socially differentiated partially because institutional parameters determine the underlying vulnerability and adaptive capacity of societies. A recent study by Houston et al. (2021) concluded that vulnerability to flooding was sensitive to financial resources and institutional support. Furthermore, a study in Kenya by Wairimu Ng'ang'a and Crane (2020) asserted that adaptation to climate change is a contingent process highly influenced by wealth, age, and gender, because these factors shape the positions from which actors negotiate, engage in, and evaluate different technical adaptation practices.

Lee and Juteau (2003) developed the theory of social differentiation in an open analytical sense, which could explain the patterns and processes behind the categories and hierarchisation of social differences and differentiation. They did so by viewing social differentiation as a product of social inequality as well as an instrument for its reproduction. Hence, this approach pays attention to political processes, norms, and representations, and to social policies; more specifically, to how they shape economic functioning and relate to normative definitions.

FRG fragmentation is already a clustering process, meaning that there is hierarchical and sectoral differentiation in existing institutional functions and processes. Therefore, Lee and Juteau's (2003) patterns and processes of social differentiation are appropriate to understand the types of FRG fragmentation in Kenya, including in Tana River County, its norms and processes, and its consequences for the floodplains. The theory then offers the opportunity to reflect on the patterns via which the processes of FRG differentiation affect the laymen, the recipients of FRM. Lee and Juteau (2003) set out these patterns, related to age, gender, ethnicity/race, and region/location. This paper reveals how these patterns echo the processes of fragmented FRG and attempts to analyse if any other patterns of social differentiation emerge from the area under review.

1.2 | Methodology

1.2.1 | Study context

Tana River County is the study area in this paper (see Figure 1). It contains a large part of the lower catchment of Kenya's longest and most expansive river catchment, the Tana River, and the Tana River Delta (Geeraert et al., 2015; Leauthaud et al., 2017). Flood inundation canals and floodplain agriculture are implemented complementarily in the



FIGURE 1 Tana River County location. Source: Okoko (2022b).

study area by Giriama, Goshi, and Pokomo ethnic communities and by pastoralist communities, Orma, Somali, and Wardey, as well as by the Wata, hunter gatherers (Okoko, 2022a, 2022b; Zenebe et al., 2022). These communities both depend on floodplain ecosystem services and contend with the destructive impacts of floods, which promote and disrupt their livelihoods (Terer, Ndiritu, and Gichuki, 2004; Leauthaud et al., 2013).

Kenya does not have a law that governs DRR; it is still a Senate bill (Government of Kenya, 2021). However, there has been a National Policy for Disaster Management (NPDM) in Kenya since March 2009, which was formulated to institutionalise mechanisms for addressing disasters (Government of Kenya, 2009). The NPDM mandates the Ministry of Special Programmes in the Office of the President to coordinate DRR initiatives within a unified policy framework in an initiative-taking manner at all levels. This includes the establishment and strengthening of disaster management institutions and partnerships, networking, and mainstreaming DRR in the development process to strengthen the resilience of vulnerable groups to cope with potential disasters. However, this document preceded the SFDRR and devolution to a county governance system; an amendment to reflect this reality is absent.

In 2013, the National Disaster Management Unit (NDMU) was established by a Presidential Directive as an effective and competent disaster management body with an established command structure, budget, and standard operating procedures (SOPs) based on best practices (Government of Kenya, 2024). Together with the Ministry of Interior and Coordination of National Government, the NDMU developed a National Emergency Response Plan (NERP) and SOPs (Government of Kenya, 2024) and the National Disaster Manual – Basic Incident Management Guidelines in 2014. Additional national policy documents for DRR include the Food and Nutrition Security Policy of 2011 and its Implementation Framework of 2018.

On 30 October 2020, nine months after the completion of this study, the Tana River County Disaster Risk Management Act, No. 6 of 2020 passed into law. Later, on 23 May 2022, Tana River County unveiled its Disaster Risk Management Policy (DRMP).

1.2.2 | Methods

Formal interview sampling applied the snowball approach (Babbie, 2008; Ritchie, 2015). At the beginning of fieldwork, a formal interview at the National Disaster Operations Centre (NDOC) in Nairobi made known the government institutions, non-governmental organisations (NGOs), and other actors involved in providing flood security in Tana River County. The formal interview with the NDOC revealed that all Tana River County government departments, national government line ministries, NGOs, and community-based organisations (CBOs) are part of the DRR steering group coordinated by the Ministry of Interior and Coordination of National Government. Officers in these ministries, departments, and organisations comprised the key informants; interviewing one led to information on how to contact others.

The study sampled the following agencies and NGOs for formal interviews: World Food Programme (WFP); the Food and Agriculture Organization of the United Nations (FAO); the United Nations Children's Fund (UNICEF); the Kenya Red Cross Society (KRCs); Concern Worldwide; Nature Kenya; and Maridhiano CBO.

The study sampled the following national government institutions: State Department for Economic Planning; Tana River County Education Office; National Drought Management Authority (NDMA); National Environment Management Authority (NEMA); Water Resources Authority (WRA); Ministry of Health (MoH); and the Kenya Wildlife Service (KWS). The Tana River County government offices sampled included Gender, Sports, Youth and Culture, Land Use Planning, Water, Environment and Mineral Resources, and Special Programmes. Another sampled office was the National Assembly Office, Garsen Constituency. Furthermore, the study sampled the Tana and Athi Rivers Development Authority (TARDA), a regional government entity charged with the management and use of trans-county resources of the Athi and Tana Rivers.

The interviews took place at organisational offices and lasted between 20 and 40 minutes. The topics discussed included FRG and other functions of the organisation or department, the challenges faced because of diversification

Table 1 shows the respondent organisations and the title and biological sex of the participants.

Respondent organisation	Title of respondent	Biological sex of respondent (M = male; F = female)
NDOC	Director of Disaster Operations	M
NDMA	Resilience Officer	M
	Field Officer	F
Nature Kenya	Programme Manager	M
	Secretary	F
	Extension Officer	M
Concern Worldwide	Project Manager	F
	Field Officer	M
UNICEF	Nutrition Officer	M
WFP	Programme Manager	M
Maridhiano CBO	Project Manager	M
County Urban Planning Ministry	Chief Officer	F
Directorate of Education, Tana River County	County Director	M
TARDA, Tana Delta Irrigation Project	Project Manager	M
	Assistant Project Manager	M
WRA	Project Manager	M
Ministry of Agriculture	County Director	M
Livestock Department	County Officer	M
County Water Department	Director	M
County Gender and Social Services Department	Director	F
County Ministry of Special Programmes	Director	M
County Ministry of Environment and Mineral Resources	Director	M
MoH, Garsen	Public Health Officer	M
	Clinical Officer	F
NEMA	Officer	F
KRCS	Community Chairperson, Tana River County	M
KWS	Warden, Tana Delta	M
FAO	Project Officer, Tana River County	M
State Department for Economic Planning	Director	M
Kenya National Assembly	Member for Garsen	M

Source: author.

This study triangulated formal interview data from state and non-state agencies with a document review of the legal framework governing FRM in Kenya, including Tana River County. The national policy documents selected for review were the NPDM, NERP and its SOPs, the National Disaster Manual – Basic Incident Management Guidelines, and the Sessional Paper No. 8 of 2012 on the National Policy for the Sustainable Development of Northern Kenya and other Arid Lands (hereafter ASAL Policy). County documents reviewed were the Tana River County Disaster Management Act and the Tana River County's DRMP. All consist of strategies, structures, and institutions for FRM and are vital for understanding the reproduction of socio-spatial inequalities and hence vulnerabilities in the case study area.

Interview questions spanned the functions of the actor in question in Tana River County, the specific FRG functions of the organisation, the challenges it faces in implementing its functions during floods, how the organisation works with other entities involved in FRG in Tana River County, and the challenges posed by this cooperation, the FRM gaps confronted by the organisation, and their causes. Lastly, there was a question on the socio-spatial vulnerabilities encountered in Tana River County while implementing programmes, the vulnerable groups or areas, and the particular challenges to implementing programmes for specific groups or in specific areas.

1.2.3 | Materials and data analysis

The materials analysed for this study are legal and policy documents governing FRM in Kenya and specifically Tana River County (see the study context subsection above), as well as transcripts of audio and written formal interviews. The main codebook for analysis (see Table 2) shows the principal themes of segment extraction from these resources. They include situations of fragmentation in FRG functions in Tana River County, the challenges this presents for FRM strategy decisions and implementation, the resultant FRM gaps, and socio-spatial inequalities in relation to the subthemes of age, gender, ethnicity, and regionality, based on Lee and Juteau (2003). The analysis interprets coded segments subjectively and explains them using patterns and processes of social differentiation.

1.2.4 | Ethical considerations

This study obtained approval from the Norwegian Centre for Research Data to conduct fieldwork on human subjects. It also obtained a National Commission for Science, Technology, and Innovation permit to carry out a study with human subjects in Kenya. An additional approval was acquired from the Ministry of Interior and Coordination of National Government, Tana River County, because it is the coordinator of the County Steering Group on DRR and the authority that gave permission to interview national government officers in Tana River County. The author also obtained approval from Tana River's Office of the County Secretary for interviews with county government officers.

TABLE 2 Codebook.

Name	Description
Challenges of fragmentation	Issues resulting from fragmentation of FRM function areas.
Diversification	FRM roles and function areas of different actors in Tana River County.
FRM gaps	Unaddressed FRM challenges in the Tana River floodplains.
Socio-spatial inequalities	Differentiation by age, gender, ethnicity, and regionality.

Source: author.

1.3 | Results

This section exposes the study's findings on how fragmented functions in FRG affect and keep reproducing socio-spatial inequalities and vulnerabilities to floods in the Tana River floodplains, in line with the coded themes. The first subsection looks at the history of diversified disaster risk governance in Kenya and issues surrounding how diversified FRG still exists in FRM even after the promulgation of the National Disaster Management Policy (NDMP) in 2009, which contained strategies for an integrated approach to disaster risk management. In addition, it also illustrates the fragmented functions and roles. The second subsection explores socio-spatial inequalities among groups and locations in the Tana River floodplains. The third subsection reveals current challenges due to fragmented FRG functions and FRM gaps in policy and practice, including a focus on drought management over flood management.

1.3.1 | Diversified FRM in Kenya and Tana River County

According to the Director of Disaster Operations in Kenya, the main function of the NDOC in Nairobi is to coordinate everything to do with disaster issues and ensure that operations are legitimate, as well as to report to the SFDRR Secretariat. From the interview, it was clear that flood disaster issues are blanketed under other disasters, except for drought, and that their governance is responsive and fragmented among all actors involved in FRG, including national government line ministries, county government ministries and departments, private sector institutions, regional government authorities, international and local NGOs, CBOs, and faith-based organisations (FBOs).

A review of the NDMP reveals that the policy did not lead to diversified FRG. In fact, it calls for integrated management of disasters and mainstreaming in development processes. From the time of the NDMP's promulgation in 2009 to the flagship of the NDMU, the NERP, and the SOPs in 2013 and 2014, there was a transitional vacuum in FRG in Kenya. First, the policy design was a centre-periphery system and could not be easily translated into a devolved system of government without an amendment. In addition, the National Disaster Management Act was not in place—it remains a bill. Consequently, the system of fragmentation of FRG functions filled the transitional vacuum and became the norm, dominated by flood emergency rescue and response activities. In addition, there was not, and there still is not, a flood management policy or authority in Kenya.

NDOC correspondence indicated that in other floodplains, such as the River Nzoia floodplains, synonymous with the Budalangi floods of 2007, the WRA applied its integrated water resource management framework to manage floods with the community. The system was successful, with the WRA coordinating resettlement, flood defences, the emergency warning system, and floodplain land use for agricultural production and water supply.

However, in Tana River, NDOC revealed that FRG was a function shared not just by national and county government departments, but also by other stakeholders. For instance, KenGen (the electricity generation company managing the Seven Forks scheme dams in the upper catchment of the Tana River), TARDA, international and local NGOs, and CBOs and FBOs all had specific projects in Tana River County.

Drought management, however, is different from other disasters because Kenya Vision 2030 and its ASAL Policy recognises drought and institutes a special system to govern it. The NDMA addresses drought risk with clear structures, funding, and response and monitoring mechanisms. In arid and semi-arid lands such as Tana River County, a focus on drought in policy encourages dryland farming close to settlements or in riverine/low-lying areas for local food security. Floods and droughts both call for food security interventions. However, it is clear from the development agenda of Kenya, fragmentation in food security strategies for droughts and other disasters, such as floods, leads to conflicting disaster risk management functions. Drought management policy encourages people to settle and farm on riverbanks for food security; an action that puts them at risk during floods.

In Tana River County, actors and strategies differentiate the situations surrounding FRG fragmentation, as shown in Figure 2. According to the formal interviews, there are seven main FRG areas in Tana River County: flood monitoring; water course and water resource management; flood prevention; emergency warning services;

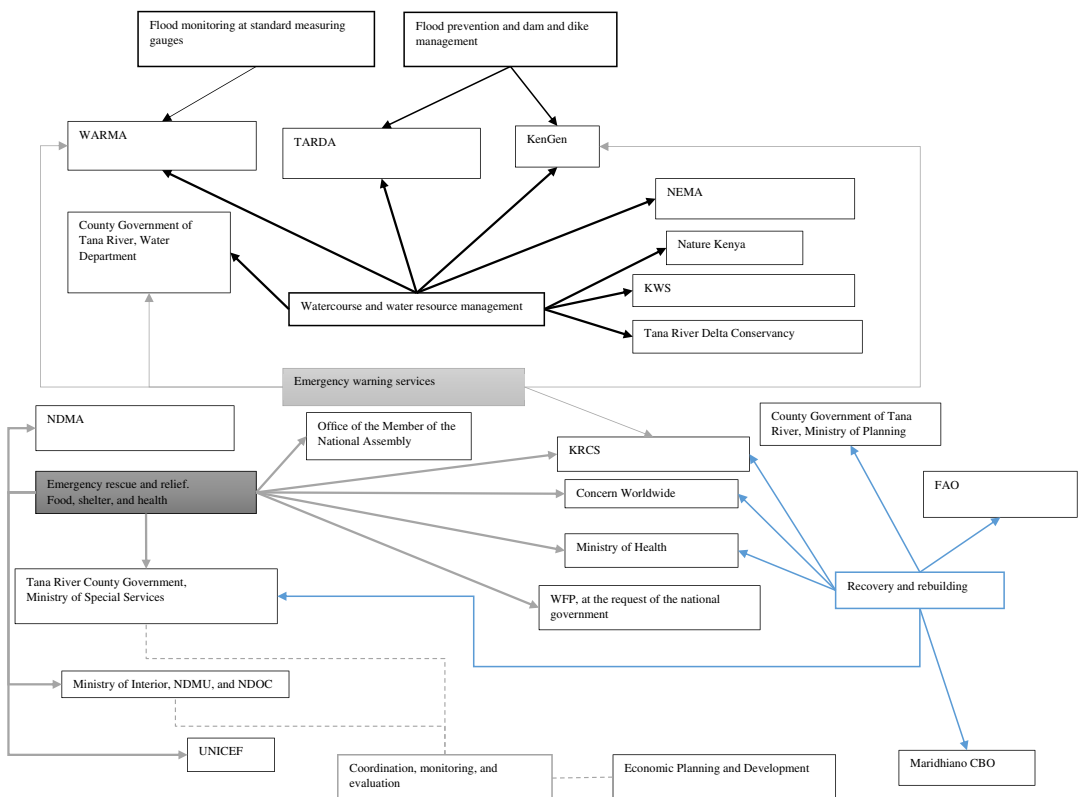


FIGURE 2 Functional differentiation of FRM strategies in Tana River County, according to document reviews and formal interviews. **Source:** author.

emergency rescue and relief; coordination, monitoring, and evaluation; and recovery and rebuilding. In Figure 2, the black arrows link to institutions involved in governing flood prevention and defence, which includes flood monitoring at the standard measuring gauges, dam and dike management, and watercourse and water resources management. The grey arrows link to institutions governing emergency response functions, which include emergency warning systems and emergency rescue and relief. The blue arrows link to institutions governing recovery and rebuilding functions, whereas the dashed lines link to institutions governing coordination, monitoring, and evaluation of flood emergency response, rebuilding, and recovery interventions. Figure 2 clearly shows fragmentation of functions both among and within the different institutions.

Divergent functions within these strategies are the responsibilities of the ministries, departments, authorities, and organisations working within FRG. For example, in relation to flood prevention, KenGen manages upstream dams while TARDA is responsible for managing the rubber dam, dikes, and embankments downstream. Therefore, KenGen can claim that floodwaters entering the Tana River catchment beyond the dams are not its responsibility, whereas TARDA only has responsibility for dikes and dams at irrigation schemes on the river, not in village settlements outside of irrigation schemes. In the case of rescue and relief, for instance, there is a limited number of boats staffed by authorities like TARDA, the Office of the Member of the National Assembly for Garsen Constituency, and the KRCS; among the three, only the KRCS has trained divers. Hence, if medical staff arrive for emergency rescue, they must wait for one of these boats to be available and for the KRCS to avail rescue volunteers. Furthermore, public health officers can only provide information to evacuees at camps because donors directly fund NGOs. Moreover, the Ministry of Interior and Coordination of National Government coordinates funding for flood risk management,

which is disseminated through national government line ministries. This limits the funds and expertise available to the county government for FRM. Such disconnections in FRG create delays that make those who need flood protection, emergency warning, rescue, and relief more vulnerable, as revealed in the subsections below on the challenges of fragmentation, FRM gaps, and resultant socio-spatial inequalities.

1.3.2 | Challenges of fragmentation and FRM gaps

The formal interviews revealed that fragmentation in FRG presents challenges, as elaborated in Figure 3. These consist of the planning- and implementation-related challenges in black boxes, the information-related challenges in grey boxes, and the actual preparedness and response consequences in light grey boxes. The challenges on the planning and implementation side of FRG directly affect data and information functions, which in turn affect the reception of FRM strategies at the local level. The results showed that both the data and information and the planning and implementation challenges owing to a fragmented FRG function affect flood preparedness and hence the flood emergency response, reinforcing and producing the FRM gaps in Table 3. The ramifications include poor reception of flood emergency warnings and resettlement strategies at the local (community) level, donor autonomy in terms of choice of relief and relief recipients, unavoidable loss of life and property due to a delayed response, neglected groups and sectors, and flood issues.

These challenges contribute directly or indirectly to the FRM gaps in Table 3. The FRM gaps align with flood preparedness and flood emergency rescue and response consequences illustrated in Figure 3. The latter shows that

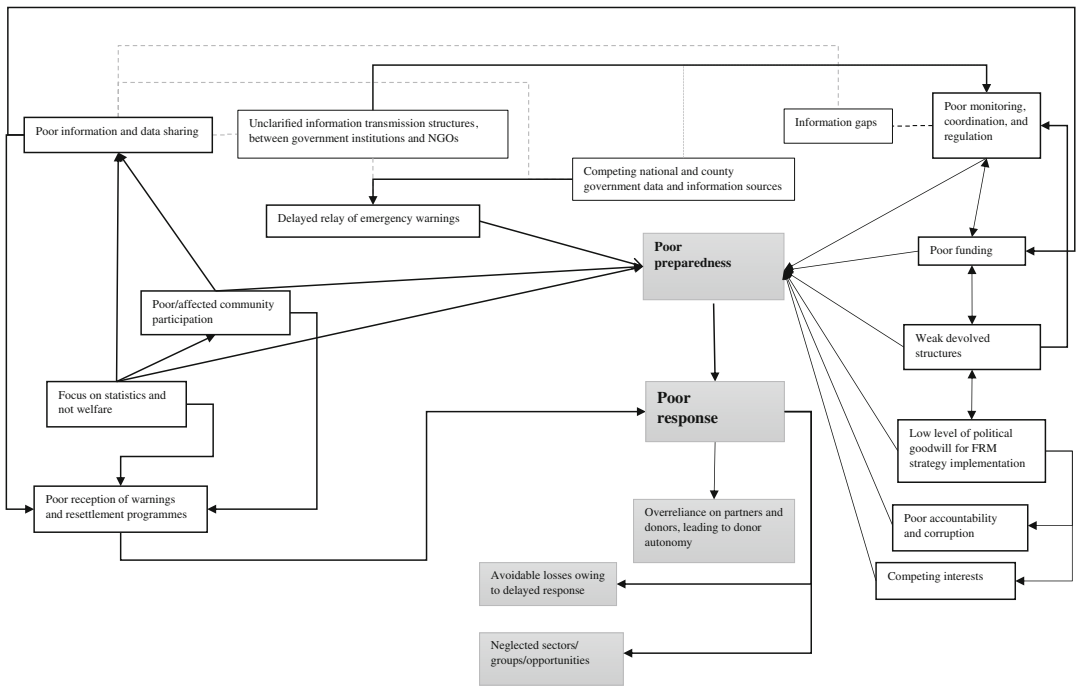


FIGURE 3 Data, information, planning, and implementation challenges due to fragmented FRG functions in Tana River County and their consequences for FRM strategy reception, flood preparedness, and response. **Note:** the black outlined boxes show the challenges of fragmentation in FRG functions and the arrows show the direction of direct consequences owing to them, either creating more challenges or affecting flood preparedness and response. The dashed lines link to specific issues of poor information and data sharing challenges. **Source:** author.

TABLE 3 FRM gaps in Tana River County.

Flood preparedness gaps	Specific limitations	Specific limitations in flood emergency rescue, response, and recovery gaps
Strategies for people living with disabilities (PLWDs)	Funding for transportation during rescue.	A lack of transportation for PLWDs.
Livelihoods strategies	No community safety nets. Inadequate livelihood safety nets. Poor information to farmers on appropriate flood crops. A lack of diversity in economic practices.	
Water and sanitation	Insufficient tap water supply. Insufficient staff for medical outreach services.	Insufficient water purifiers.
Flood protection	Poor dike maintenance and weak dikes. Poor management of riparian zones. Riverbank protection. Loss prevention. A lack of multipurpose dams to enable irrigation.	
Land use	Uncontrolled riverbank settlements. Inadequate land adjudication. No demarcation of the boundaries to block people from putting structures along riverbanks.	
Flood water resource management	No floodwater harvesting and supply infrastructure.	
Relief food security and standardisation	Poor regulation of donor items.	Donors provide relief food without diversity for a balanced diet. Donors flood the relief kitty with formula milk that is inappropriate in flood emergencies because of poor water quality and access to fuel. In addition, it interferes with the promotion of breastfeeding.
A lack of long-term FRM strategies	No flood disaster management fund. No flood disaster response centre. No long-term food security strategy for flooding.	Floods have become an endemic problem. Most of the activities are focused on immediate emergency interventions.
Tidal FRM strategies	No alternative agricultural activities.	Salinisation by tidal floods during drought.
	A lack of a regulatory reservoir to ensure that high tide matches the level of water in the river.	
Recovery and rebuilding	No permanent solution to settle the ever displaced.	
Transportation sector	Poor roads.	Marooned villages and schools cannot access relief aid, children cannot go to school, and the sick cannot access hospitals.

TABLE 3 (Continued)

Flood preparedness gaps	Specific limitations	Specific limitations in flood emergency rescue, response, and recovery gaps
Capacity development	Insufficient medical and public health personnel. Insufficient training for flood rescue and response personnel. Limited medical supply reserves.	Avoidable loss of life during rescue.
Inclusion of cultural issues	No inclusion of cultural issues in flood emergency response planning.	Poor uptake of water and sanitation interventions, such as mobile or temporary toilets.
A lack of interventions in schools	No flood protection for school buildings. Poor schoolchildren participation in FRM strategy development. Poor inclusion of schoolchildren's issues in FRM strategies.	Schoolchildren's needs during floods are not addressed, such as the need for more text and exercise books, food, and toilets.

Source: author.

limitations in flood preparedness are similar among the factors of land use and flood protection (including in schools). Responses for livelihood strategies are like those for a lack of long-term FRM strategies; and one solution to them lies in floodwater resource management. Other gaps include those that hinder rescue, relief, and recovery, such as gaps in the transportation sector, water and sanitation, relief food security and standardisation, recovery and rebuilding, tidal FRM strategies, and capacity development, with respect to the inclusion of cultural issues and a lack of interventions in schools. All of these gaps individually and collectively reinforce existing and reproduce new socio-spatial inequalities, as presented in the subsection below.

1.3.3 | Socio-spatial inequalities

The results pointed to five patterns of socio-spatial inequalities, related to: age; gender; ethnicity; regionality; and disability. Age-related patterns of socio-spatial vulnerability were higher among children and the elderly. Rescue of the latter was a challenge because available rescue boats and personnel are deficient of the accommodations they require for boarding and transit. Among children, current FRM strategies do not safeguard rights and access to education services. For example, lorries ferry school feeding programme food, but they cannot cross floodwaters, leaving the children who can get to school to study on an empty stomach. In addition, there was no means of safe transportation from marooned villages to school, putting children at risk of drowning and wildlife attacks. There was also no allocation to replace books, uniforms, backpacks, and other stationery swept away by floodwaters.

Gender-related patterns were higher among women, especially among pregnant and lactating mothers. For instance, there were no enclosed bath areas in evacuee camps for women, who could not bathe in the open like men. Furthermore, there were no safe breastfeeding areas or special tents for expectant and nursing women. Moreover, while donors flooded the relief kitty with formula milk, there was no effort to provide fuel and clean water to allow safe use of the formula milk to nurse infants. There was also no priority rescue tailored to pregnant and nursing women, because of the scarcity of rescue boats.

Ethnicity-related issues concerned the cultural livelihood activities of groups inhabiting the floodplains. In this case, farmers were more vulnerable because unlike pastoralists, they could not move their crops to protect them

from the floods, and while there are safety nets for pastoralists (through the Ministry of Agriculture and Livestock Development and the NDMA), such as insurance compensation and supplementary feeds, there are none for farming communities, except relief food. This makes it difficult for them to recover their livelihoods after floods.

Region-related vulnerabilities concern the exposure of settlements to flood hazards. For example, marooned villages were more vulnerable because aid could not reach them, and floodwaters submerged their farms. Also, farming communities in tidal floodplains were the victims of a neglected flood hazard: tidal flood salinisation. This means that there is no available safety net or FRM strategy for them in terms of water, food, and livelihood security, making them more vulnerable.

An emerging pattern of socio-spatial vulnerability in this paper was disability-related. First there is poor representation of people living with disabilities and human immunodeficiency virus/acquired immunodeficiency syndrome (HIV/AIDS) in FRG. This cascades towards unaddressed issues in rescue and response. First, there is no rescue manual for people with different disabilities during floods and no disability-enabled rescue boats. Second, there is no priority rescue for those with disabilities and serious mobility challenges, such as those in wheelchairs, on crutches, and the visually impaired. Third, there are no strategies to immediately replace devices swept away by floods, such as hearing aids. Fourth, there are no rescuers specifically trained to assess the issues of people living with disabilities. Fifth, the evacuee camps, including the mobile toilets, are inaccessible by evacuees with disabilities.

The results reveal that spatial inequalities reproduced a link to ignored groups and sectors in the FRM strategies implemented within the fragmented systems, while the produced inequalities establish a link to the FRM gaps resulting from the presence of a fragmented system. For example, poor preparedness puts age- and gender-differentiated populations such as children, the elderly, and expectant and lactating women at more risk because they require more resources to evacuate and settle in evacuee camps. Even those sheltering in place still require more assistance, to gain access to schools, healthcare facilities, and nutritional services. Poor monitoring of flood emergency preparedness, response, recovery, and rebuilding links to poor learning and assessment of gaps that differentiate people by age, gender, culture, and region, making them more vulnerable to losses and suffering, and even death, in future flood emergencies. This in turn makes addressing emergencies more expensive with every flood event because the fragmented system is unable to produce long-term solutions and retains a focus on relief and rescue, which then become increasingly expensive. This creates a spiral of overlooking inequalities, not learning and sharing information, repeating the same mistakes, having funding deficiencies, reproducing existing differences (vulnerabilities and inequalities), and producing new forms of differences, trapping floodplain residents in a never-ending vulnerability web.

2 | DISCUSSION

Not all diversified FRG results in detrimental fragmentation, though, especially where it is intentional, and policy-led, as in cases in Europe. In such instances, a policy provides the template for amendment and clear, coordinated institutions within which rectification of competency and responsibilisation issues is possible. This case study, however, presents one of the least studied scenarios of unintended or unplanned fragmentation in FRG functions, especially in Sub-Saharan Africa. Leck et al. (2018) deduced that in Sub-Saharan Africa, fragmented governance, donor priorities, and inadequate monitoring of hazards, impacts, and vulnerability constrain effective FRM. In this study, unintended fragmented governance is the nexus of inadequate monitoring of hazards, impacts, and vulnerability. This cascades into poor planning, financing, and allocation and coordination of competencies and responsibilities, affecting flood emergency response in terms of delays, avoidable losses, and donor autonomy.

The processes of FRG decision-making and cooperation in the results reveal that while the role of diversified government policies and agencies is to manage floods, they could produce fragmented responsibility without clear coordination, which features entrenched bureaucracy and competition. This is similar to observations made on

fragmented FRG in Africa by Adekola et al. (2020) in Calabar, Nigeria, and Okunola and Simatele (2022) and Okunola et al. (2022) in southern Nigeria and South Africa.

A further exacerbating factor in FRG processes established in this study could be the presence of too many actors with competing interests. For instance, in the case of Tana River County, there are two levels of government, county and national, TARDA and KenGen govern the Tana River watercourse, and the NGOs and CBOs based in Tana River County are responsible for drought and flood response, conflict management, and livelihoods empowerment, all factors related to flooding in the area. In addition, fragmented policy development processes could produce conflicting policy actions, such as the food security policy action for Tana River County, with its arid and semi-arid land, which encourages settlement and cultivation on riverbanks. This heightens risk in times of flooding by increasing the exposure of such settlers, reinforcing the region/location differentiated socio-spatial patterns of vulnerability. Moreover, a lack of a specific national or regional FRM policy that specifies SOPs and incidence management guidelines could make the FRG actors autonomous and unaccountable. This produces data sharing, financing, staffing, community participation, and cultural incorporation deficiencies in FRG functions and decision-making processes, which undermine FRM strategies, especially in emergency response.

Consequently, the challenges of fragmentation and the resultant FRM gaps keep reproducing and producing socially differentiated socio-spatial inequalities and vulnerabilities among floodplain residents in Tana River County. These inequalities align with Lee and Juteau's (2003) patterns of social differentiation because they are either related to age, gender, ethnicity, culture, and regionality.

Addressing age- and gender-related patterns of vulnerability among women in FRG is important because in African countries most women lack access to land, livestock, and financial capital, which are critical in reducing vulnerabilities to disasters (Chineka et al., 2019). A study in Dhaka, Bangladesh, pointed out that women, infants, and young children are the groups most vulnerable to flood impacts, both during the flood emergency and in its aftermath (Goudet, 2010; Goudet et al., 2023). In Tana River County, the differentiation is not just between but also within groups. Among women there are women-headed households, expectant women, lactating women, and women living with disabilities and/or HIV/AIDS. Among children, the differentiation is between infants and school-going children, each vulnerable for varied reasons. In addition, recent studies show that elderly people are among the groups most vulnerable to the negative impacts of disasters (Bayraktar et al., 2018). Another study concluded that the psychological and physiological health effects of floods appear to be borne disproportionately by females, the elderly, and children during floods (Lowe, Ebi, and Forsberg, 2013). Therefore, generalisation of relief aid, as well as a lack of post-flood strategies, increases the impact of floods on these groups and their vulnerability, which at times could mean loss of life.

This study deduces that ethnicity and spatial-related differences in vulnerability interconnect because of the culture of people, where they live, and their livelihood activity not being separate. This pattern of differentiation intricately links to loss of property, interruption of livelihoods, poor access to roads and markets, and poverty. From the study results in Table 3, one can deduce that FRG decisions in Tana River County do not give enough emphasis to the culture and traditions of the residents of the Tana River floodplains. Thus, at the community level, there is resistance to FRM strategies such as resettlement plans. In addition, the fragmentation challenge of poor information sharing has led to distrust of warnings among floodplain residents; most people choose to shelter in place until they can see the floodwater rising. This is a clear illustration of the disconnect between experts and laymen, as pointed out by Clausen (1992).

Disability-related differentiation, meanwhile, stems from a lack of representation in FRG decisions and functions and a lack of rescue guidelines and relief aid tailored to people with disabilities. These issues are unaddressed because no actor in the FRG system in Tana River County is specifically responsible for people living with disabilities in floodplains. The differentiation also has other differentiations within it, based on the type and severity of the disability: more vulnerable groups of children with disabilities, expectant and lactating women with disabilities, and elderly people with disabilities.

This study recommends implementation of the current NDMP because it sets out to address the challenges of fragmentation in FRG functions, the resultant FRM gaps, and the produced and reproduced socio-spatial inequalities and vulnerabilities; and it should have an amendment to accommodate a devolved system of governance. This would ensure that integration or diversification of FRG functions is intentional, and policy-led, and therefore it is easy to configure exact points of fragmentation in FRG functions, institutional challenges, and implementation gaps and address them effectively. Implementation of the NDMP should include the development and enactment of the National Disaster Management Act, which is not yet in place, and a clear implementation framework for disaster management in Kenya. The framework should comprise specific disaster management policies for each disaster, similar to the ASAL Policy, the inclusion of these policies in the country's development agenda, Vision 2030, to ensure financing, the creation of relevant institutions, and staffing and equipping of these institutions with competencies and capacity to monitor and reduce risk, and prepare for, respond to, and recover from specific disasters in their realm. These institutions should establish a way to coordinate with county and regional authorities and other actors on FRG at local levels, including floodplain residents, through information sharing, the deployment of staff and equipment, and planning, monitoring, and evaluation, to prevent the FRM gaps that materialised in this study.

To tackle the socio-spatial differences established here, the study recommends that FRM policies include a clause that ensures representation of these groups in institutions and in FRG decision-making processes, especially for emergency response. The exacerbation of the socio-spatial vulnerabilities in this study, especially those related to age and gender, could just be a beginning. Involving these groups will enable them to present and address all of their needs pertaining to flood response and recovery and avoid overlooking them, as in the current fragmented FRG system and functions. In addition, since most socio-spatial differences in vulnerability increase because the FRG system overlooks interventions in the aftermath of floods, there is need for a clear flood recovery strategy, with relevant financing, rebuilding, oversight, and public participation.

This study also recommends the establishment of permanent rescue centres in the floodplains, equipped with necessities for all groups, as well as wheelchair access and child-friendly boats and rescue vehicles. In the event of a flood, floodplain residents, especially those in marooned and hard-to-reach areas, can then shelter in place without worrying about provisions. It also recommends protection dikes in the floodplains as well as raised permanent housing, where practical and possible. Governance actors could also empower and enable local communities to plant trees along riverbanks, as well as flood crops as buffers and for food security. At governance levels, there is a need for oversight of private actors, such as KenGen in Kenya, which manages and uses the river catchment and claims to prevent floods to ensure that its actions are as flood control-based as they are profit-based. The same oversight should apply to donor agencies and NGOs to ensure that their actions do not aggravate socio-spatial vulnerabilities. Also of benefit would be a clear emergency warning system that is timely and accurate regardless of which FRG actor relays information to the public, to foster trust in FRG institutions and the adoption of FRM strategies by floodplain populations.

3 | CONCLUSION

This study set out to answer the following research question: 'What are the consequential socio-spatial vulnerabilities of fragmentation in FRG functions in Tana River County, Kenya?'. The literature from Europe revealed that extensive scholarship exists on fragmentation of FRG functions as part of intended diversification of FRG and limited studies exist on unintended diversification of FRG, which this paper addresses. The scenario of unintended fragmentation of FRG functions in the study area, Tana River County, exists because of the delayed implementation of the existing NDMP. This policy set out to tackle the challenges of fragmented FRG functions, such as funding, monitoring, accountability, competing interests, data and information sharing, politics, preparedness, avoidable losses of life, and donor issues. This study revealed a link between these challenges and FRM gaps. The latter pointed to FRG actors overlooking socio-spatial inequalities, a process that facilitated their reproduction or the production of new

inequalities. Patterns that differentiate the inequalities are related to age, gender, ethnicity/culture, regionality/location and/or disability.

The recommendations of this study include process- and pattern-oriented points. In the decision-making processes of FRG, it is vital to implement the NDMP and to enact the relevant act of parliament to guide its institutionalisation. It is also important to create disaster-specific policies, aligned institutions, implementation frameworks, and funding and monitoring mechanisms. Another key step is to empower and enable the participation of socio-spatially vulnerable groups in FRG decision-making processes and oversee private and non-governmental actors. Since this is only one case of unintended diversification of FRG leading to unintended fragmentation of FRG functions, this study also recommends conducting more studies of this kind, especially in Sub-Saharan Africa.

ETHICS STATEMENT

This paper reports analysis of primary data. The ethics of data collection and analysis were approved by the Norwegian Centre for Research Data and the National Commission for Science, Technology, and Innovation, Kenya. Persons from whom data were collected gave their free, prior, and informed consent for the collection and analysis of information. Their data have been kept confidential and used anonymously to protect their identity.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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ENDNOTE

¹ For more information, see <https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030> (last accessed on 29 April 2024).

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